KING SAUD UNIVERSITY
Deanship of Common First Year
Department of Basic Sciences

STAT 101
Introduction to Probability and Statistics

Semester: First
Year: 1448H
Number of Pages: 4

The First Paper Homework

Deadline: Monday, 28/9/2026

Student Name: ..... **ID:** ..... **Section:** .....

Question 1: Select the correct answer for each of the following:

1) A class has a frequency of 12 and an ascending cumulative percentage frequency of 75% in a data set containing 40 observations. How many observations are below the lower boundary of that class?

a) 16 b) 18 c) 20 d) 22

2) A set of data contains 90 observations. The ascending cumulative frequency for a particular class is 68. How many observations are above the upper boundary of that class?

a) 18 b) 22 c) 68 d) 90

3) Six observations have a mean of 12. When the largest observation is removed, the mean of the remaining five observations becomes 10. What is the largest observation?

a) 18 b) 20 c) 22 d) 24

4) A statistician wants a measure that divides an ordered set of data into four equal parts. Which measure should be used?

a) Percentile b) Decile c) Quartile d) Mean

5) An ordered data set contains 19 observations. A percentile calculation produces the rank $p_r = 12.6$. The 12th and 13th observations are 74 and 81. Find the percentile value.

a) 74 b) 77.5 c) 78.2 d) 81

6) A set of data contains 39 ordered observations. Find the rank of the 8th decile.

a) 30 b) 32 c) 31.5 d) 33

7) For which of the following sample sizes does the calculated rank of the 75th percentile correspond exactly to an ordered observation?

a) $n = 18$ b) $n = 19$ c) $n = 20$ d) $n = 22$

8) A grouped data set consists of 6 classes. The midpoint of the first class is 8, and the class length is 5. What is the range of the data?

a) 20 b) 25 c) 28 d) 35

9) If $Q_1 = 20$ and $IQR = 12$, what is Q_3?

a) 28 b) 30 c) 32 d) 34

10) If the standard deviation remains constant while the mean increases, the coefficient of variation (CV) will generally

a) Increase b) Decrease c) Remain unchanged d) Become negative

11) A statistical report includes an ACFP constructed over five continuous classes. The plotted points are (0,0), (10,5), (20,7), (30,20), (40,36), and (50,50). What is the relative frequency of the third class interval?

a) 0.14 b) 0.20 c) 0.26 d) 0.50

12) A category has a relative frequency of 0.275. What is its percent frequency?

a) 0.275% b) 2.75% c) 27.5% d) 275%

13) Category A has a relative frequency of 0.18, while Categories B, C, and D have frequencies of 27, 37, and 18, respectively. What is the total number of observations?

a) 80 b) 90 c) 95 d) 100

14) When constructing a frequency distribution table, the data have a smallest value of 12 and a greatest value of 47, and the data are divided into 5 classes. What is the calculated class boundary length?

a) 90 b) 7.0 c) 7.2 d 150

15) The mean of five numbers is 18. If four of the numbers are 12, 15, 20, and 25, what is the value of the fifth number?

a) 16 b) 18 c) 20 d) 23

16) Class A consists of 20 students whose scores have a mean of 70, and Class B consists of 30 students whose scores have a mean of 80. What is the mean of the scores for all 50 students?

a) 74 b) 75 c) 76 d) 78

17) Which measure is identical to the median?

a) P_{25} b) D_2 c) Q_2 d) Q_3

18) Which z-score represents the observation closest to the mean?

a) -2.4 b) 1.8 c) -0.2 d) 1.1

19) Which formula correctly calculates a z-score?

a) $\frac{x_i-\bar{x}}{S}$ b) $\frac{\bar{x}-x_i}{S}$ c) $\frac{x_i+\bar{x}}{S}$ d) $\frac{x_i-\bar{x}}{S^2}$

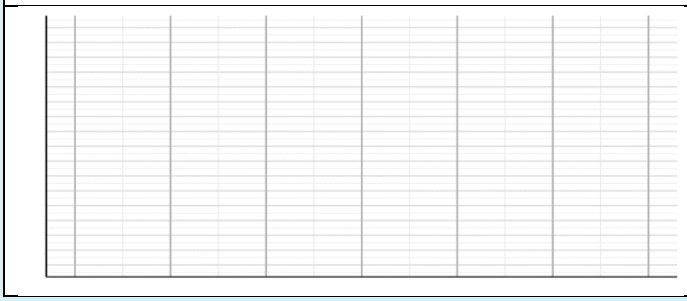
20) Which situation regarding the value x would produce a negative z-score?

a) $x > \bar{x}$ b) $x < \bar{x}$ c) $x = \bar{x}$ d) $S = 0$

Question 2: Answer the following statements with True or False.		**Answer**
1	Two sets of data with the same mean must have the same standard deviation.	.....
2	A z-score of zero means that the observation is equal to the mean.	.....
3	An observation greater than the higher fence is classified as an extreme value.	.....
4	It is possible for the mean, median, and mode of a set of data to be equal.	.....
5	If the range of a set of data is zero, then the IQR must also be zero.	.....
6	A variance of zero means that all observations are equal.	.....

Question 3: A university department conducted a survey to evaluate enrollment distribution across four major academic specializations by gender (male vs. female).

a. Represent the given data in a multiple bar chart.



Academic Specialization	Male Students	Female Students	Total
(1): Data Science	12	14	26
(2): Software Engineering	22	16	38
(3): Cybersecurity	16	14	30
(4): Artificial Intelligence	20	6	26
Total	70	50	120

Question 4: A continuous set of data contains 96 observations. The minimum value is 17 and the maximum value is 64. The researcher decides to use 6 classes.

1. Calculate the range.

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2. Determine the class boundary length.

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3. Construct the class limits for all classes.

.....

4. Compute the class midpoint for the last class.

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Question 5: Consider the following three data sets.

Data set X	5	10	15	20	25	Note
Data set Y	10	20	30	40	50	$y_i = 2x_i$
Data set Z	20	40	60	80	100	$z_i = 2y_i = 4x_i$

1. Calculate the mean for each of these data sets. Find the relationship between the three means.

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2. Calculate the standard deviation for each of these data sets. Find the relationship between the three standard deviations.

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Question 6: Consider the following ascending cumulative frequency polygon (less than ogive) of grouped data, representing the weight to 30 fruit boxes:

1. How many boxes have a weight of at least 20 kg?

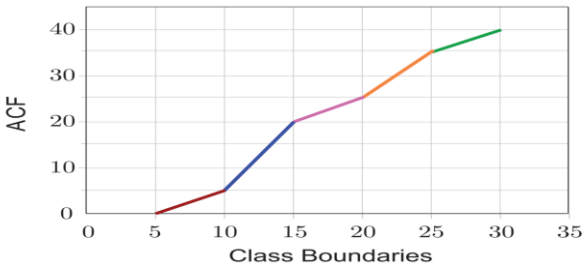
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2. What percentage of the fruit boxes weigh less than 20 kg?

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3. Calculate the mean.

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4. Calculate the median.

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5. Represent the above data using a polygon

Question 7: The following set of data represents the test scores of 15 students in an introductory statistics quiz:

12 15 16 18 18 20 21 22 24 25 26 28 30 32 50

Then

1. Calculate the mean.

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2. Calculate the median.

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3. Identify any outlier in the dataset.

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4. Construct the box plot for the given data.

Question 8: The following table shows the number of books read by a group of students during a month.

1. Calculate the mean.

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2. Calculate the median.

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Number of Books	Frequency
0	4
1	7
2	12
3	10
4	12

The End– Good Luck